

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc. I.T Semester V)
Applied Business Analytics
PRACTICAL NO - 7

Roll No.: T018	Name: Sachin Ramesh Mehta
Class: TYIT	Batch: 1
Date of Assignment: 03-10-26	Date/Time of Submission: 03-10-26

7. Sampling and Sampling Distributions:

a. Conduct a survey and collect data from a sample population, ensuring proper sampling techniques are employed.

CODE:-

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Load CSV file
df = pd.read_csv("Waste Management Survey Responses - Form Responses 1.csv")

# Display first five records
print("First Five Records:")
print(df.head())

# Display number of responses
print("\nTotal Number of Responses:", len(df))

# Select numerical variable
# -----

#Satisfaction with waste management
column = "9. How satisfied are you with waste management in your area?"

satisfaction = pd.to_numeric(df[column], errors="coerce").dropna()

# Calculate statistics
sample_size = len(satisfaction)
mean_value = satisfaction.mean()
median_value = satisfaction.median()
std_value = satisfaction.std()

print("\nStatistics:")
print("Sample Size:", sample_size)
print("Mean:", round(mean_value, 2))
print("Median:", round(median_value, 2))
print("Standard Deviation:", round(std_value, 2))

# Display frequency of satisfaction ratings
print("\nSatisfaction Frequency:")
print(satisfaction.value_counts().sort_index())

# Create suitable chart
plt.figure(figsize=(7,5))
```

```
satisfaction.value_counts().sort_index().plot(
    kind="bar"
)

plt.title("Waste Management Satisfaction Survey")
plt.xlabel("Satisfaction Rating")
plt.ylabel("Number of Respondents")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```

OUTPUT:-

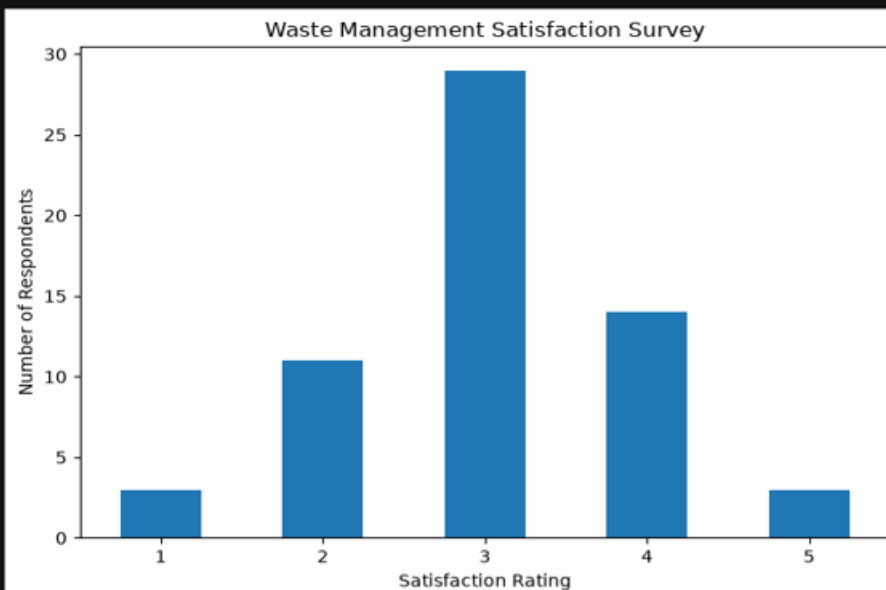
```
First Five Records:
      Timestamp      1. Name \
0  9/19/2026 22:30:54      Azma
1  9/19/2026 22:45:29  Leena Mondal
2  9/19/2026 22:55:44 Santosh prajapati
3  9/19/2026 23:08:32      Radhika
4  9/19/2026 23:14:59      Sid

      2. Which area/locality do you live in? \
0      Santacruz (east)
1      Bhayandar
2      Naigaon east
3      Andheri East
4      Santacruz/ Andheri

      3. How many people are there in your household? \
0      52
1      4
2      180
3      4
4      3
```

```
      4. Approximately how much waste does your household generate per day? (in kg) \
0      3/4 kg
1      1kg
...
3      29
4      14
5      3
Name: count, dtype: int64
```

Output is truncated. View as a [scrollable element](#) or open in a [text editor](#). Adjust cell output [settings...](#)



b. Use the Central Limit Theorem to analyze the sampling distribution of a sample mean and estimate population parameters.

CODE:-

```
# Select random sample of 30 observations
```

```
np.random.seed(42)
```

```
sample_30 = satisfaction.sample(  
    n=30,  
    random_state=42  
)
```

```
print("\nRandom Sample of 30:")  
print(sample_30)
```

```
# Calculate mean of selected sample  
sample_mean = sample_30.mean()
```

```
print("\nMean of Selected Sample:", round(sample_mean, 2))
```

```
# Repeat sampling 100 times
```

```
sample_means = []
```

```
for i in range(100):
```

```
    random_sample = satisfaction.sample(  
        n=30,  
        replace=False  
    )
```

```
    mean = random_sample.mean()
```

```
    sample_means.append(mean)
```

```
# Store sample means
```

```
sample_means_df = pd.DataFrame({  
    "Sample Number": range(1, 101),  
    "Sample Mean": sample_means  
})
```

```
# Display first 10 sample means
```

```
print("\nFirst 10 Sample Means:")  
print(sample_means_df.head(10))
```

```
# Mean of all sample means
```

```
mean_of_sample_means = np.mean(sample_means)
```

```
print(  
    "\nMean of All Sample Means:",  
    round(mean_of_sample_means, 3)  
)
```

```
# Standard deviation of sample means
```

```
sd_sample_means = np.std(  
    sample_means,  
    ddof=1
```

```

)

print(
    "Standard Deviation of Sample Means:",
    round(sd_sample_means, 3)
)

# Histogram of sampling distribution
plt.figure(figsize=(7,5))

plt.hist(
    sample_means,
    bins=10,
    edgecolor="black"
)

plt.title(
    "Sampling Distribution of Sample Means (n = 30)"
)

plt.xlabel("Sample Mean")
plt.ylabel("Frequency")

plt.tight_layout()
plt.show()

# Standard Error
standard_error = std_value / np.sqrt(30)

print(
    "\nStandard Error:",
    round(standard_error, 3)
)

# Compare original mean and sampling mean
print("\nComparison:")
print(
    "Original Sample Mean:",
    round(mean_value, 3)
)

print(
    "Mean of Sampling Distribution:",
    round(mean_of_sample_means, 3)
)

# Sampling distributions for different sizes
sample_sizes = [10, 20, 30, 40]

results = []

for n in sample_sizes:

    means = []

```

```

for i in range(100):

    random_sample = satisfaction.sample(
        n=n,
        replace=False
    )

    means.append(random_sample.mean())

results.append({
    "Sample Size": n,
    "Mean of Sample Means": np.mean(means),
    "SD of Sample Means": np.std(
        means,
        ddof=1
    )
})

comparison = pd.DataFrame(results)

print("\nComparison of Different Sample Sizes:")
print(comparison.round(3))

# Estimate population mean
population_mean_estimate = mean_value

print(
    "\nEstimated Population Mean:",
    round(population_mean_estimate, 2)
)

# Save sample means to CSV
sample_means_df.to_csv(
    "Sampling_Distribution_Sample_Means.csv",
    index=False
)

print(
    "\nSample means saved successfully!"
)

# Conclusion
print("\nConclusion:")
print(
    "The sampling distribution of the sample means is centered "
    "close to the original sample mean. As the sample size "
    "increases, the variation among sample means decreases. "
    "This demonstrates the Central Limit Theorem and shows "
    "that the sample mean can be used to estimate the "
    "population mean."
)

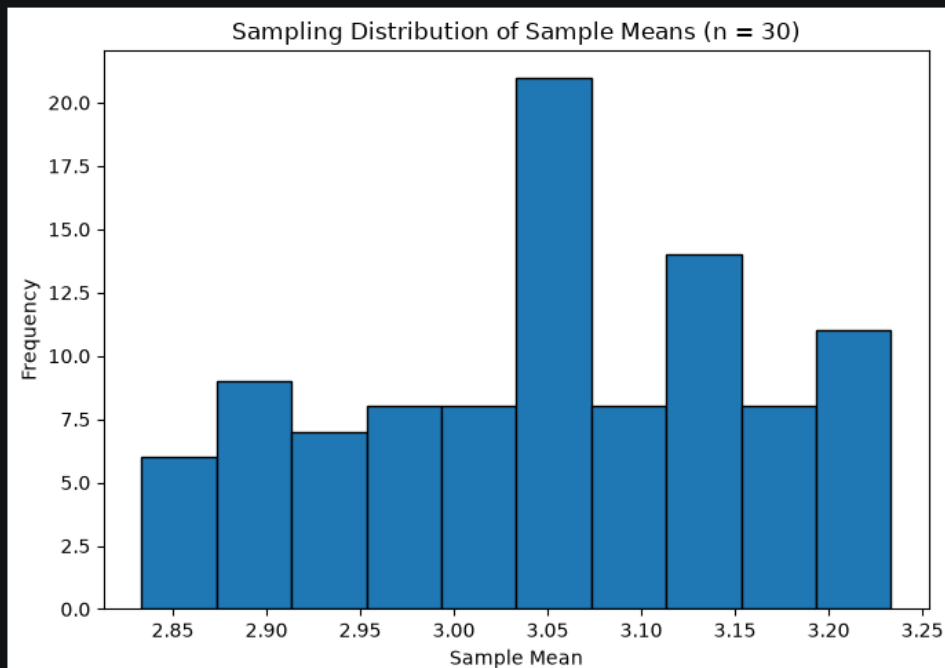
```

OUTPUT:-

```
Random Sample of 30:
0      2
5      4
36     5
45     3
13     2
54     1
33     3
48     4
12     3
57     2
46     3
50     3
31     3
3      3
52     3
17     2
8      4
6      3
40     4
4      3
43     1
19     2
34     4
...
9      10      2.900000

Mean of All Sample Means: 3.049
Standard Deviation of Sample Means: 0.105
```

Output is truncated. View as a [scrollable element](#) or open in a [text editor](#). Adjust cell output [settings...](#)



Standard Error: 0.166

Comparison:

Original Sample Mean: 3.05

Mean of Sampling Distribution: 3.049

Comparison of Different Sample Sizes:

	Sample Size	Mean of Sample Means	SD of Sample Means
0	10	3.052	0.255
1	20	3.046	0.160
2	30	3.049	0.130
3	40	3.042	0.080

Estimated Population Mean: 3.05

Sample means saved successfully!

Conclusion:

The sampling distribution of the sample means is centered close to the original sample mean. As the sample size